



Implementing RMM 1.6: Boeing's Experience

Agenda

- Our Puget Sound Data Centers
- The Initial Conversion to RMM
- The Hybrid
- Implementing RMM 1.6
- Gotcha's
- Where are we now?
- Questions

Our Puget Sound Data Centers

- 3 Locations (separated by 1,800 miles)
- 500,000 physical tapes
- 300,000 virtual tapes
- 4,000 – 4,500 write mounts/day
- Between 500 and 1,000 vault tapes per day
- z/OS, VM, Linux, Windows, etc.
- Batch interface supports non-z/OS nodes
- STK silos for all platforms
- 29 nodes spread over 6 tape complexes
- Using 3480, 3490, 3490E, 9840, 9940, LTO, LTO-2, 3590, 4/8-MM

Our Puget Sound Data Centers (cont.)

- Plex 1 – Operating System test plex (4 nodes)
 - 3 nodes a true SYSPLEX
 - 4th node shares minimal DASD, silo, virtual box
- Plex 2 – Development plex (2 nodes), a true SYSPLEX
- Plex 3 – General purpose plex (6 nodes)
 - All nodes share silo and a virtual box
 - None share RACF, HSM, or catalog
- Plex 4 – High availability plex (4 nodes), a true SYSPLEX
- Plex 5 – Engineering and modeling (7 nodes)
 - All nodes share silo – no virtual box
 - 4 separate HSM plexes, catalogs, RACF databases

Our Puget Sound Data Centers (cont.)

- Plex 6 (6 nodes) – VM plex
 - Share silo, no virtual box
 - Separate from the rest of the enterprise (except for batch interface)
 - Volumes move to and from z/OS plexes
- Distributed servers – (8, at present)
 - Separate silos, no virtual box
 - Separate from the rest of the enterprise (except for batch interface)

The Conversion to RMM

- The decision: One RMM CDS for the enterprise
 - A hybrid of TLCS and RMM
 - Retain library names from TLCS
 - All silos and racks in the same physical location
 - Stay in record mode (testing in WARN and PROTECT mode was problematic)
 - Training, training, training
 - Over 3 shifts, leaving someone to work in the library
 - Learn new field names and commands
 - Process changes

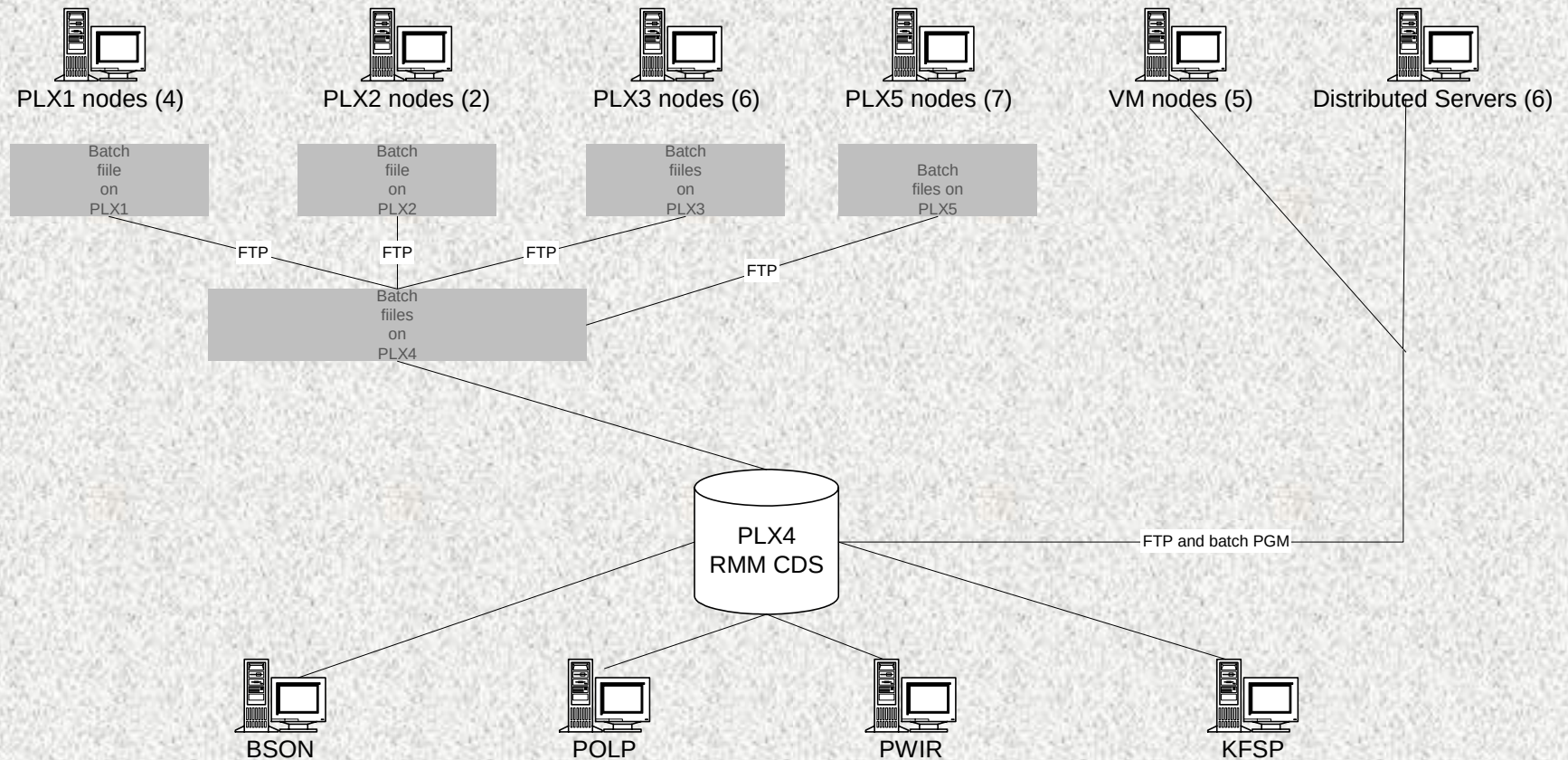
The Conversion to RMM (cont.)

Facility class name	Access for RMMPSUPR	Access for RMMPLIBR	Access for RMMPOPER	Access for RMMUSER
STGADMIN.EDG.FORCE	Update	None	None	None
STGADMIN.EDG.HOUSEKEEP	Read	None	None	None
STGADMIN.EDG.LISTCONTROL	Control	Control	Control	Control
STGADMIN.EDG.MASTER	Control	Update	Read	Read
STGADMIN.EDG.OPERATOR	Update	Update	None	None
STGADMIN.EDG.RELEASE	Read	Read	None	None
STGADMIN.EDG.RESET.SSI	Alter	None	None	None
STGADMIN.EDG.VRS	Control	Control	Read	Read

The Hybrid

- High availability plex (PLX4) uses vanilla RMM
- Remote nodes used SMF records and FTP to build RMM updates
 - Requires multiple started tasks
- Interface built for VM, UNIX/LINUX, Windows
 - Provide JCL and batch program
- Dialog interface resembles TLCS
 - Allows for processing multiple volumes simultaneously
 - Achieve customer buy-in

The Hybrid, Pre RMM 1.6



TLCS/Hybrid Dialog interface (cont.)

```
..... Tape Volume Transactions .....  
:  
:  
: OPTION: V      A - Add          E - Evaluate   K - Recall    R - Release  
:  - Del VRS     B - Bin view    G - Gum label M - Move      U - Update  
: # - Generate   C - Certify    H - Hold      N - New       V - View  
: " - Reload     D - Delete    I - Check in  O - Check out Y - Yield bin  
:  
:  
: >-----+-----+-----+-----+-----+-----+-----+-----<  
: | G05932 G18767 J06535 J18366 J29949 J30491 J30855 J83754 J92837 J94463 |  
: | J95041 J95194 L05949 L09163 L17601 L72860                               |  
: >-----+-----+-----+-----+-----+-----+-----+-----<  
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```

TLCS/Hybrid Dialog interface

..... Tape Volume Table Display Row 1 to 19 of 20

> CUR :
:
:.....:

Vol	Stat	Avail	Home	Loc	Bin	Job name	Expdt	Data set name
G05932	M	VR	Z	0		PACRO123	05132	P.ACROXXXX.VSAMTST1.G3
G18767	M	VR	Z	0		PACRO6E0	05128	P.ACROXXXX.KGPEDSS1.G3
J06535	S		Z	Z		HPDM		TSSM.NODE2.SYSDUMPS.W9S
J18366	M	VR	Z	0		PACROED5	05152	P.ACRO0000.TESTVAL1
J29949	M	VR	Z	0		PACROD90	05132	P.ACROXXXX.HSTRYDUP
J30491	M	VR	Z	0		PACROK10	07057	P.ACRONODE.DEVLFULL.G0
J30855	M	VR	Z	0		PACROD90	05132	P.EPICXXXX.UONOFDUP
J83754	M	VR	Z	0		HSM	99365	HSMMIG.COPY.HMIGTAPE.D
J92837	M	VR	Z	0		HSM	99365	HSMMIG.COPY.HMIGTAPE.D
J94463	M	VR	Z	0		HSM	99365	HSMMIG.COPY.HMIGTAPE.D
J95041	M	VR	Z	0		PACRO@00	05136	DB2XCTLG.ARCHCPY2.D051
J95194	S	PR	Z	Z		HPDM		TSSM.NODE1.DUMPDSN
L05949	S		Z	Z		PACRO\$10		P.ACROXXXX.BGA1DPC1.SY
L09163	S	PR	Z	Z		PACRO\$10		P.ACROXXXX.BGA1DPC1.SY
L17601	S		Z	Z		PACROB10		P.DUMPTOOL.PKAWC052.G0
L72860	S	OL	Z	Z		PACRO@00		DP30CTLG.DP34.ARCP2.D0

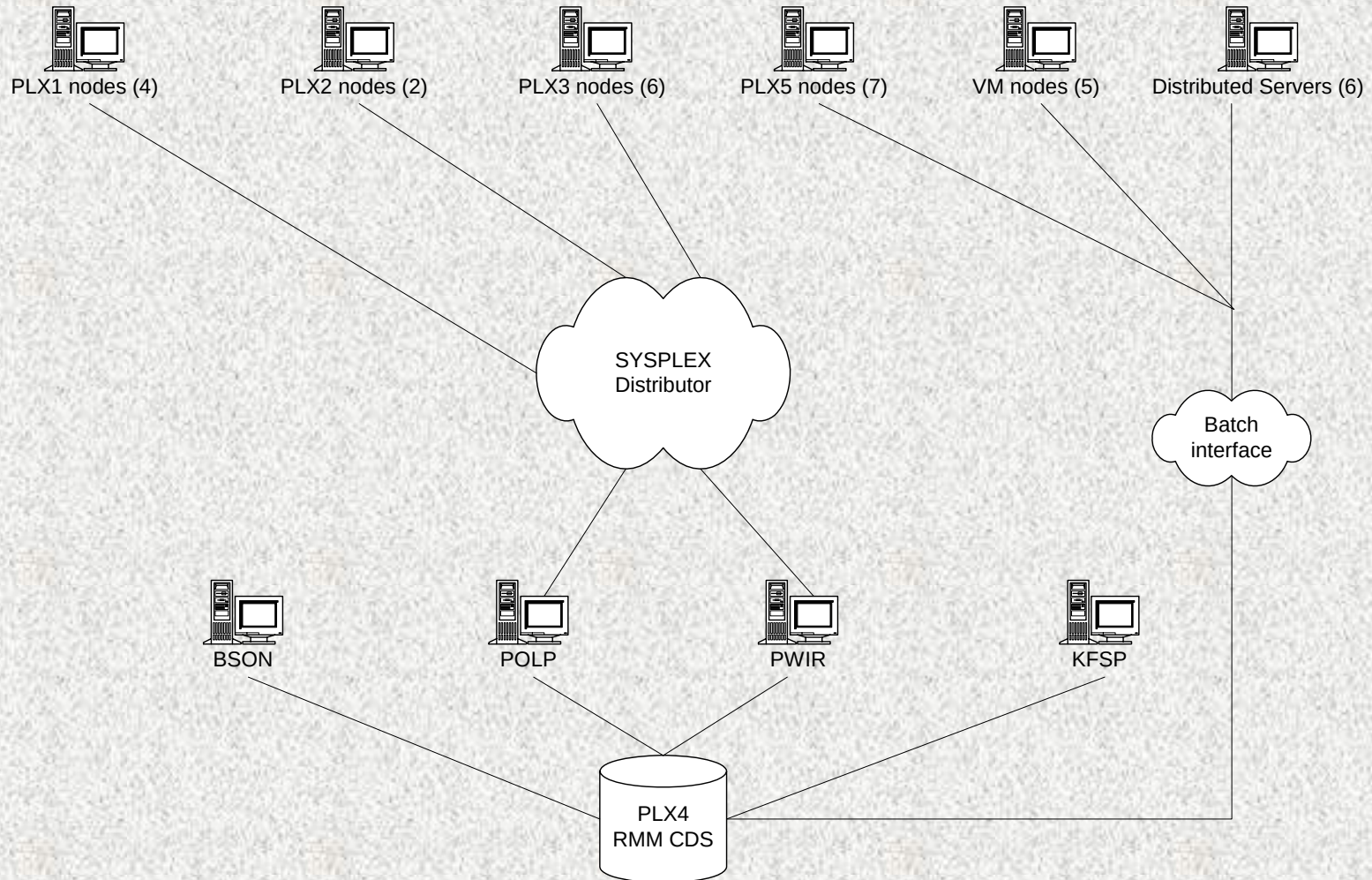
Implementing RMM 1.6

- Defining user ID's and RACF
- Building TCP/IP SYSPLEX distributor
- PLX4: Two servers, two stand-alone systems
- Fixing the main CDS
 - Run MEND to recover from 1,800 errors
- The SLSUX06 exit
 - Need a new way to track inserts/ejects
- Automation updates and migration
 - RMM now needs to start/recover automatically on all nodes

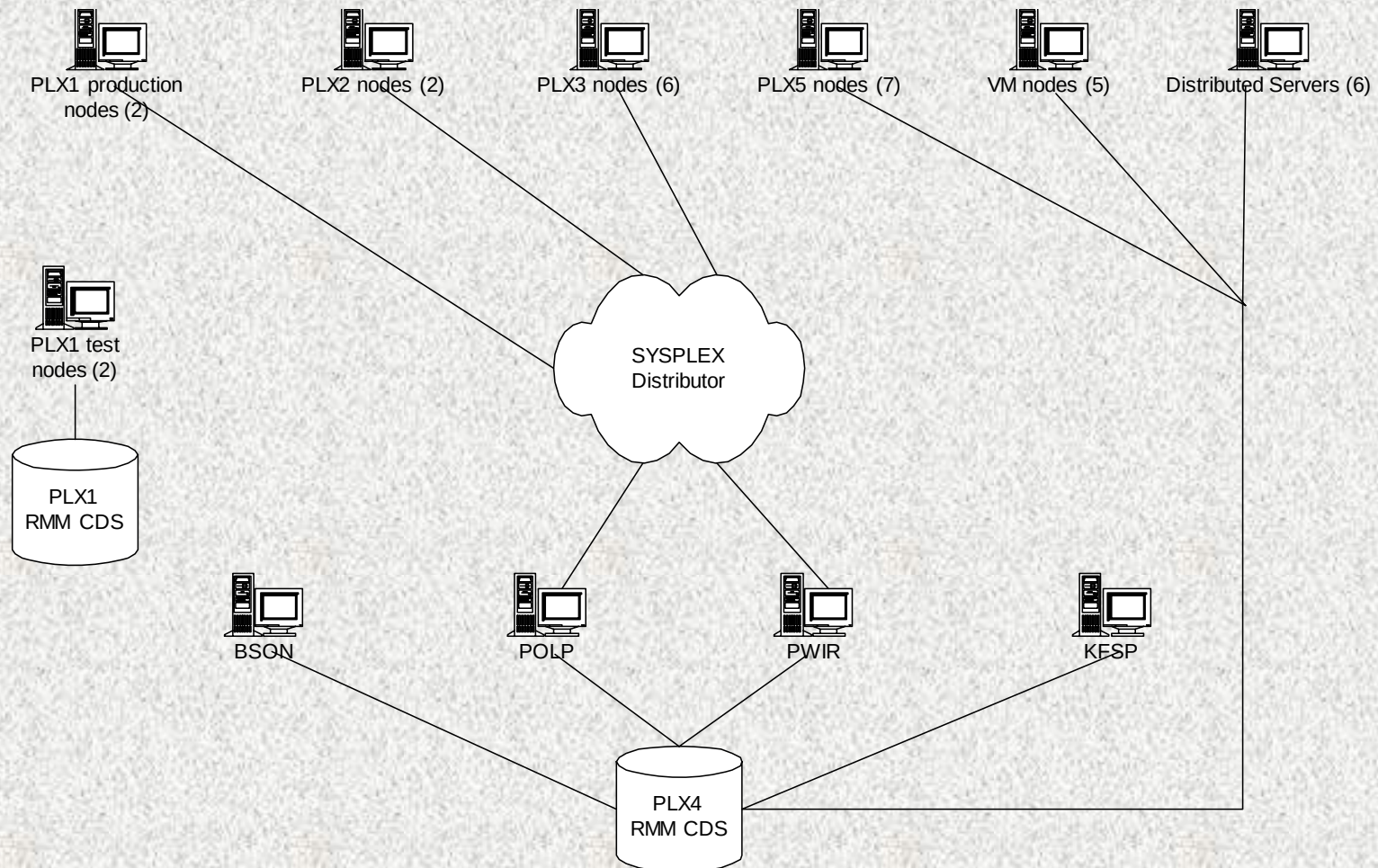
Implementing RMM 1.6 (cont.)

- PLX5 doesn't want to play
 - Systems “going away” to distributed world
 - Back-leveled and won't take maintenance
 - Have to keep the home-grown started tasks around a while longer
 - Still have to do relabels
- 5 PARMLIB members (prod server, prod client, test server, test client, stand-alone)
- Isolating a test Environment
 - Minimize risk to production environment
 - Build processes for testing new software
 - Multiple housekeeping runs???

System layout under RMM 1.6



System layout under RMM 1.6 with test environment



SYSPLEX Distributor Defintion

; DFRMM

; POLP is the primary distributor. POLP and PWIR are targets.

;

VIPADEFINE MOVEABLE IMMEDIATE distrip1 distrip2

VIPADISTRIBUTE distrip2 PORT port#

DESTIP destip1 destip2

;

; DFRMM Back-Up Definition (POLP is the Primary Distributor

; POLP and PWIR are targets.)

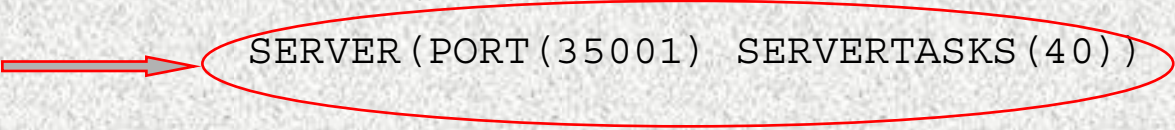
;

VIPABACKUP 30 distrip2

;

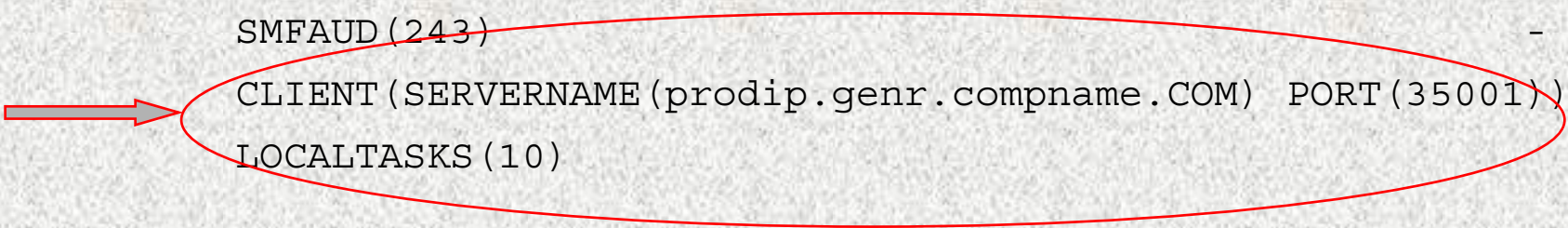
PARMLIB definition for Server

```
/* **** */
/*                                           */
/* Production server PARMLIB member          */
/*                                           */
/*-----*/
OPTION  DSNAME(RMMP.MASTER.CDS)      /* master file name */ -
        JRNLNAME(RMMP.RMM.JOURNAL) /* JOURNAL NAME */   -
        VRSCHANGE(INFO)              -
        BLP(NORMM)                    -
        VRSEL(NEW)                     -
        OPMODE(P)                      -
        SMFAUD(243)                    -
        SERVER(PORT(35001) SERVERTASKS(40))
```



PARMLIB definition for Client

```
/******  
/*  
/* Production Client PARMLIB member  
/*  
/*-----*/  
OPTION  VRCHANGE (INFO) -  
        BLP (NORMM) -  
        VRSEL (NEW) -  
        OPMODE (P) -  
        SMFAUD (243) -  
        CLIENT (SERVERNAME (prodip.genr.compname.COM) PORT (35001)) -  
        LOCALTASKS (10)
```



PARMLIB definition for Stand-alones

OPTION	DSNAME (RMMP.MASTER.CDS)	/* master file name */	-
	JRNLNNAME (RMMP.RMM.JOURNAL)	/* JOURNAL NAME */	-
	VRCHANGE (INFO)		-
	BLP (NORMM)		-
	VRSEL (NEW)		-
	OPMODE (P)		-
	SMFAUD (243)		

Gotcha's

- Journal back-up, CDS reorgs, CDS VERIFY/MEND's
 - Do you need them?
 - Schedule them, do not do them ad hoc
- Force option on commands
 - Understand the implications of the commands!
- Interface with STK's HSC/VSM
- Relabels
- RACF replication
- Processes for testing new software
- Tivoli Storage Manager tapes not getting released

Where are we now?

- Deleting lots of old volumes (hence, the need for periodic reorgs)
- RMM running without “help” on all nodes except PLX5
- TLCS/Hybrid dialog interface still place
- Inserts/ejects still tracked via SMF/FTP
- All PLEX's in WARN mode, moving to PROTECT mode
 - Takes advantage of HSC/VSM interface
 - Eliminates some inits/erases
- Manually synchronizing location/status of PLX5 tapes
- Distributed server batch interface maturing
- Training, training, training
- Documentation of the implementation
- Process changes

Questions?

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